INCH-POUND
MIL-DTL-55302/52D
w/AMENDMENT 1
12 September 2012
SUPERSEDING
MIL-DTL-55302/52D
17 December 2003

DETAIL SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: RECEPTACLE, STRAIGHT-THRU, HERMAPHRODITIC CONTACT FOR PRINTED WIRING BOARDS (.100 SPACING)

This specification is approved for used by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-55302.

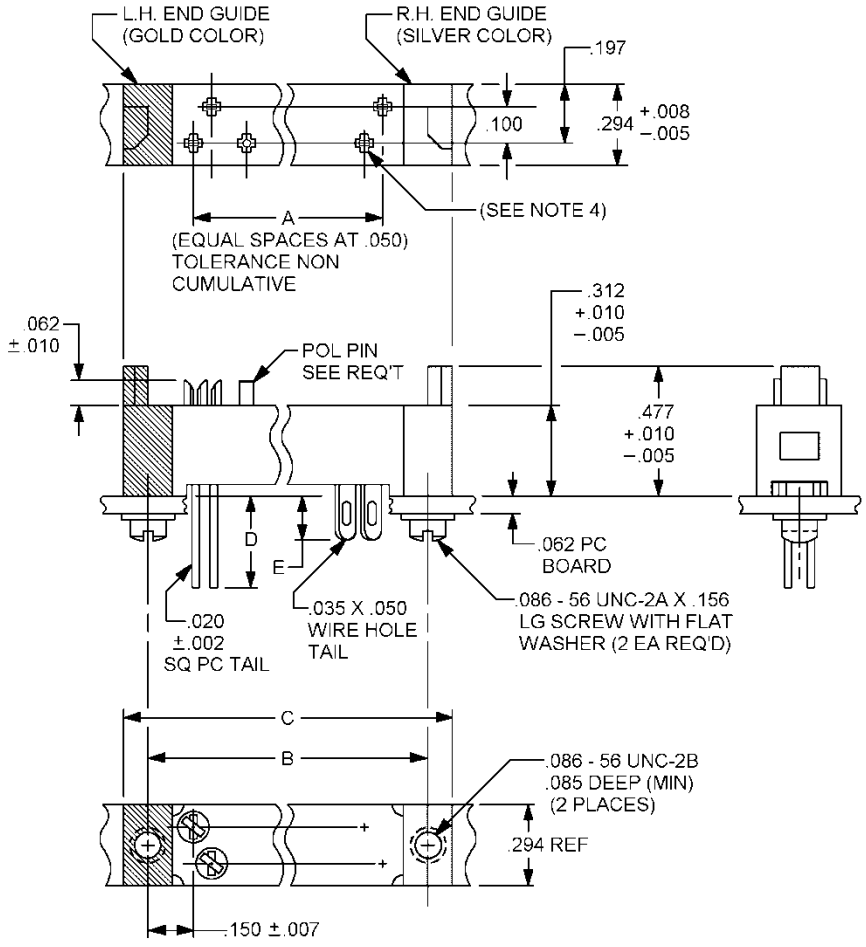
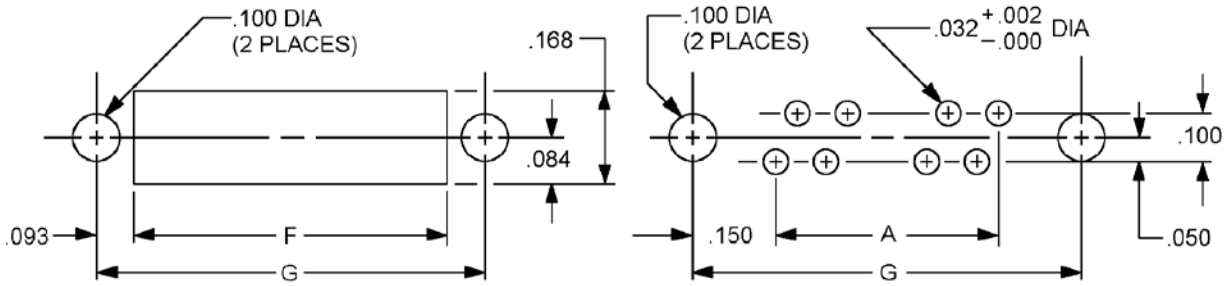


FIGURE 1. Connectors, receptacle, .100 (2.54 mm) spacing.

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RECOMMENDED CUTOUT FROR RECEPTACLE USING WIRE WHOLE CONTACTS, IN NON-CONDUCTIVE CHASSIS

RECOMMENDED HOLE PATTERN FOR RECEPTACLE USING PRINTED CIRCUIT TAIL CONTACTS

Inches	mm	Inches	mm	Inches	mm
.002	0.05	.035	0.89	.100	2.54
.005	0.13	.050	1.27	.150	3.81
.007	0.18	.052	1.32	.168	4.27
.008	0.20	.084	2.13	.197	5.00
.010	0.25	.085	2.16	.294	7.47
.020	0.51	.086	2.18	.312	7.92
.032	0.81	.093	2.36	.477	12.12

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm).
4. With connectors fully engaged the mating contacts shall have a minimum overlap distance of .062 inches (1.57 mm). As a minimum the contacts shall exhibit a wiping action through this distance.

FIGURE 1. Connectors, receptacle, .100 (2.54 mm) spacing - Continued.

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TABLE I. Dash number and dimensions.

Dash 1/ number	Number of contacts	Dimensions (inches)						
		A	B ±.010	C Reference	D ±.015	E ±.015	F	G
01	18	.850	1.150	1.300	---	.187	.964	1.150
02					.250	---	---	
03					.281	---	---	
04					.562	---	---	
05					.126	---	---	
06					.484	---	---	
07	30	1.450	1.750	1.900	---	.187	1.564	1.750
08					.250	---	---	
09					.281	---	---	
10					.562	---	---	
11					.126	---	---	
12					.484	---	---	
13	36	1.750	2.050	2.200	---	.187	1.864	2.050
14					.250	---	---	
15					.281	---	---	
16					.562	---	---	
17					.126	---	---	
18					.484	---	---	
19	42	2.050	2.350	2.500	---	.187	2.164	2.350
20					.250	---	---	
21					.281	---	---	
22					.562	---	---	
23					.126	---	---	
24					.484	---	---	
25	54	2.650	2.950	3.100	---	.187	2.764	2.950
26					.250	---	---	
27					.281	---	---	
28					.562	---	---	
29					.126	---	---	
30					.484	---	---	
31	72	3.550	3.850	4.000	---	.187	3.664	3.850
32					.250	---	---	
33					.281	---	---	
34					.562	---	---	
35					.126	---	---	
36					.484	---	---	

1/ See requirements for complete Part or Identifying Number (PIN) when polarization is required.

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Material and finish:

Contact: .020 inch (0.51 mm) thick phosphor bronze in accordance with ASTM B139/B139M, UNS C51000. Gold plate in accordance with ASTM B488, type II, grade D class 1.27, over nickel plating in accordance with SAE AMS-QQ-N-290, class 2, 30 to 150 microinches in the engagement area for a length of .120 inch (3.05 mm) minimum (see figure 1). The remainder of contact shall be tin lead .0001 microinch minimum in accordance with SAE AMS-P-81728 over nickel plating in accordance with SAE AMS-QQ-N-290, class 2, 30 to 150 microinches.

Insulator: ASTM D5948, type SDG-F.

End guides: Left hand, sintered brass 90/10, clear chromate in accordance with MIL-DTL-5541, class 1A (gold color). Right hand, sintered brass 90/10, nickel plate 300 microinches in accordance with MIL-DTL-14072 (silver color).

Polarizing pin: Aluminum alloy 2011-T3 with clear chromate, in accordance with MIL-DTL-5541, class 1A.

Mounting screw: Brass, cadmium plated, gold chromate finish.

Flat washer: Stainless steel.

Contact identification: Numerical sequence starting with the closest contact adjacent to the left hand end guide (gold color) shall be contact number 1; contact number 2 would be the next offset contact, etc.

Mating and unmating: The maximum insertion force, in pounds, shall not exceed a value equal to .5 times the number of contacts, and the withdrawal force, in pounds, shall be a minimum of .11 times the number of contacts and shall not exceed the measured insertion force.

Individual contact separation force:

Unplated contacts: One-ounce minimum when tested in production with gage shown on MIL-DTL-55302/97.

Plated contacts: One-ounce minimum.

Contact resistance: The average contact resistance of all contacts measured shall not exceed 0.010 ohm, and no individual contact pair shall have a resistance exceeding 0.020 ohm.

Low level circuit: The low-level circuit resistance shall not exceed 20 milliohms.

Dielectric withstanding voltage:

Sea level: 1,000 volts rms, 60 cycles AC.

High altitude: 300 volts rms, 60 cycles AC.

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Current rating, maximum: 5 amperes.

Polarizing pin: When required, one of the following codes shall be added to the dash number:

"P": Specifying location by contact number where polarizing pin shall be inserted; for example, M55302/52-01P17 (polarizing pin inserted in place of contact number 17).

"H": Specifying location by contact number where contact shall be omitted for mating; for example, M55302/52-01H17(polarizing hole has contact number 17 omitted).

Mating plug: Shall conform to MIL-DTL-55302/53, /54.

PIN: M55302/52- (dash number from table I).

Patent number 2,750,572: The Government has a royalty free license under this patent for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government.

Qualification: Qualification is not required for this specification sheet.

First article testing (FAT): FAT shall be in accordance with MIL-DTL-55302, qualification inspection.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-55302, this document references the following:

MIL-DTL-5541
MIL-DTL-14072
MIL-DTL-55302/53
MIL-DTL-55302/54
MIL-DTL-55302/97
ASTM B139/B139M
ASTM B488
ASTM D5948
SAE AMS-QQ-N-290
SAE AMS-P-81728

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2012-131)

Review activities:
Army - AR, AT, AV, MI
Navy - AS, MC, OS, SH
Air Force - 19, 99

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